Vayun Gupta

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Education

MS **Columbia University in the City of New York**, Computer Engineering Sep 2026 - Dec 2027
Coursework: SoC Design, AI-Native Computing, Data Center Processing, Systems & Networks for LLM Inference

BS **University of Illinois at Urbana-Champaign**, Computer Engineering Aug 2022 - May 2026
Edmund J. James Scholar and Member, HKN (IEEE Honors Society)

BS **University of Illinois at Urbana-Champaign**, Technology Entrepreneurship (ILEE) Aug 2022 - May 2026

Experience

Tanvir Ahmed Khan's Research Group at Columbia, Research Assistant May 2026 - Sep 2026
- Identified tracing overhead and built AutoTrace, cutting instrumentation time from 1.5 hours to 5 minutes per trace
- Implemented a Chisel API and FIRRTL transforms to automate trace generation and FireSim integration for SoC designs

University of Illinois Urbana-Champaign, ECE 110 Course Assistant Aug 2023 - Dec 2025
- Led labs and office hours for about 30 students, explaining circuit concepts and hardware debugging with oscilloscopes
- Mentored students in circuit analysis and design, providing individualized feedback on assignments and lab work

AI Acceleration & Systems Projects

Transformer Accelerator Design on ESP-SoC Sep 2026 - Present
- Designing attention accelerators using HLS and exploring parallelism, pipelining, and memory organization to achieve Pareto-optimality among throughput, latency, and resource utilization
- Integrating accelerator stages into ESP-SoC's heterogeneous platform and evaluating LUT/FF/BRAM/DSP usage

Data-Movement-Aware Accelerator Interface Sep 2026 - Present
- Evaluating latency and throughput across serial, buffered, and overlapped execution to identify interface bottlenecks
- Developing a buffered RTL interface to address data-movement bottlenecks between neural and symbolic accelerators

Heterogeneous LLM Inference System Sep 2026 - Present
- Evaluating heterogeneous LLM serving with GPU-based pre-fill and FPGA-accelerated decode against GPU-only inference
- Modelling KV-cache transfer, bandwidth, latency across varying prompt lengths, generation lengths, and concurrency

CUDA Optimization of CNN Inference Jun 2024 - Aug 2024
- Reduced combined convolution kernel time from ~85 ms to ~45 ms per 10,000-image batch at 87.14% accuracy
- Tuned FP16 storage, constant-memory weights, loop unrolling, and block sizes using Nsight Compute profiling

Computer Architecture & RTL Design Projects

RISC-V Branch Profiling Sensitivity Analysis for BOLT Sep 2026 - Present
- Investigating BOLT's sensitivity to RISC-V CTR entry count and record format using BOOM simulations
- Evaluating how CTR configurations affect profile quality and performance gains from BOLT's code layout optimizations

RTL Debugging Agent for OoO RISC-V Processor Jan 2026 - May 2026
- Built an RTL debugging agent that uses Verilator logs and waveform-trace summaries as tool calls, improving GPT-5.4's ability to fix real OoO RISC-V bugs from 6/15 to 15/15

Out-of-Order RV32IM Processor Design & Verification Aug 2025 - Dec 2025
- Designed a SystemVerilog OoO core with explicit register renaming, 64 physical registers, and a 32-entry ROB
- Separated tag-only CDB wakeup from PRF write-back and read operands after issue to reduce operand transport by 81%

FPGA-Based System to play Chess with Real-time Graphics Apr 2025 - May 2025
- Integrated a MicroBlaze C chess engine with SystemVerilog sprite rendering, USB mouse input, and HDMI output
- Encoded 64 board squares in 256 bits and transferred updates through AXI GPIO to hardware rendering logic

Skills

Programming Languages: SystemVerilog, Verilog, Chisel, Scala, C, C++, Python, Go, CUDA, RISC-V Assembly

Frameworks/Tools: Chipyard, FireSim, FireMarshal, Verilator, Vivado, Vitis, Synopsys, Linux, Git, Make, GDB

Skills: RTL design, verification, test benches, waveform debug, synthesis, timing closure, OoO CPUs, caches, AXI, SPI